Reviews and Technical Requirements

Technical Requirements

The rockets of all teams must comply with the following design constraints:

1. Stability `S > 0.1 * rocket length` and `S = 1-2 calibers`.
2. Velocity at the end of the launch rail `v > 15m/s` for a launch rail length of `4 meters`.
3. After the recovery event the descent velocity has to be between `10 m/s` and `15 m/s`.
4. Simulated altitude `H_sim <= 600m` using the AeroTech H238T solid rocket motor.
5. A mount for a 29mm AeroTech motor with 4 grains (`l ≈ 250mm`) must be provided. A more detailed sketch of the motor mount can be found here.
6. A flight computer must be used to trigger a recovery event and log flight data.
7. An engine ejection charge must be used as backup mechanism for separation.
8. The flight computer must not be open to any blackpowder or delay charge fumes.
9. Rail buttons must be used to guide the rocket along the launch rail. A sketch can be fund in the How to Rocketry page.

Preliminary Design Review

The main purpose of the PDR is to give you an early opportunity to get your rocket design on the right track. A short presentation of the rocket design is given in front of experienced Space Team members. The following is required for the review:

- OpenRocket simulation plot: height/acceleration/velocity over time and stability over time
- What is discussed during the review:
  - rocket dimensions
  - fin dimensions and how they are fixed to the rocket
  - flight computer and parachute location
  - payload type and location
  - point of separation for parachute deployment
  - manufacturing methods for components
  - any other special ideas or plans you have that are not trivial
- We ask you to provide any sketches or CAD drawings, that are necessary to discuss all of the mentioned topics, but a detailed OpenRocket design may suffice.

If the committee approves the design, you are allowed access to basic materials like carbon fiber, glass fiber (plates), 3D printing filament, screws etc. Further expenses on top of the basic FIRST-concept have to be provided by the respective group members on their own or have to be requested at the FIRST team leads. If the committee does not fully approve the design concept it does not mean the end of the project. Feedback will be given and suggestions will be made on how to resolve any possible design issues.

Critical Design Review

The goal of the CDR is to check if your final design is technically sufficient for a launch. The process will be similar to the PDR, but all of the rocket's parts and functions should be finalized in all the necessary detail. What you must prepare for the review:

- Updated OpenRocket simulation including component weights from CAD file or real weight if existent
- CAD models of all components (excluding cables, parachute lines, parachute and motor) with properly assigned materials and a complete assembly of the rocket (including screws etc.)
- Specific plans for the manufacturing, material and assembly of all components
- Launch checklist (example)
- Ideally some proof-of-concept models for non-trivial mechanisms that haven't been discussed in any workshops.

The launch checklist should include all necessary steps to prepare the rocket for the launch, starting from assembling subassemblies in the workshop in Vienna, integrating final parts like black powder charges at the launch site in Straubing and lastly any actions necessary to be taken once the rocket is on the launch rail like powering the flight computer and activating cameras. The launch checklist should serve as a tool for you to really think about the integration of all the rockets parts and

Some aspects that you should make sure to have thought of and have prepared are:

- how the motor is held in place
- how the railbuttons are mounted
- how the parachute is mounted
- that the electronics power button should be accessible on the launch rail, e.g. not directly in line with the railbuttons
- that the motor delay charge is able to eject the parachute in case of failure of other recovery systems
- there shouldn't be any big holes (> ~3mm) in the parachtue compartment and between the motor and the parachute for the pyrotechnic charges to work
- that all the non-structural components are accessible and assemblable, e.g. flight computer, power button, parachute mount
- how and where cables have to pass through the rocket

If you don't pass the CDR, you will have the opportunity to resolve any issues and be offered another opportunity to present your design. After passing the CDR, you shouldn't do any changes to your rocket without having it approved by the project leads.

Once you have passed the CDR a flight computer, kevlar cloth, battery and motor tube will be reserved for you.

Flight Readiness Review

The FRR is the final review before your rocket is cleared for its launch. A series of checks will be performed that will determine the risk of failure of the rocket.

Functionality checks during the review:

1. The motor casing can be inserted into the motor mount and be secured from falling out
2. The flight computer is safe from the recovery black powder charges and from the secondary ejection charge of the motor
3. The flight computer and battery are easy (enough) to mount
4. Coupler sections, connecting two separable body tube parts, are at least one caliber long.
5. The flight computer can be activated and deactivated from the outside, even if the rocket is already inserted into the launch rail.
6. The body tube has holes that allow the flight computer to measure the outside pressure, in order for the apogee to be detected reliably.
7. The parachute should fall out of the rocket when holding the rocket upside down avoid it getting stuck at deployment.
8. The coupler connecting separable rocket parts should have a sufficient fit. As a guideline, when holding the upper part of the rocket, the lower part should not slip off the coupler, but it should slip off with a bit of wiggling.
9. The railbuttons fit into the launch rail and the rocket moves with little resistance.
10. Parachute, parachute lines, carabiner and kevlar cloth have to be integrated into the rocket during the review.
11. The rocket has to be seperable by a blackpowder charge.

Preparation for the review:

1. The rocket has to be in a flight-ready state, including the flight computer, battery, parachute and parachute lines. The only exceptions are the motor and the recovery deployment charge.
2. An updated OpenRocket simulation has to be available. In the simulation, the rocket weight and center of mass have to be overwritten with the actual measured rocket weight and center of mass (without motor).
3. An updated flight checklist and list of required tools for the launch site
4. The simulation has to meet all technical requirements, most importantly:

    4.1 stability `S > 0.1 * rocket length`
    4.2 rail exit velocity `v(s = 4m) > 15m/s`

4.3 decent velocity `10m/s` and `15m/s`
4.4 Simulated altitude `H_sim <= 600m`

Launch Day Preparation

1. The updated OpenRocket file is uploaded to the folder given by the project leads with its name according to *TeamID_Teamname_Version*.
2. A pack list for all the necessary parts for the assembly of the rocket should be written by the teams to avoid missing any necessary tools or parts. Basic tools like hex keys, wrenches, pliers, etc. and motors and pyrotechnics will be provided by the organizational team.
3. The launch checklist, including all the necessary steps for assembly and functionality checks to execute during the launch preparation, must be finished.

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